

MARTINUS ALVIN HARYANTO

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EDUCATION

BINUS University

B.S. in Computer Science, Data Science Specialization Track

GPA: 3.96 / 4.00

Jakarta, Indonesia

2024 – 2028 (Expected)

PUBLICATIONS

“Dynamic Classifier Selection using Local Class Accuracy on Overlap Cleaned Heterogeneous Pool of Experts”

1st Author | Published | [Paper and Source Code](#)

Sept 2026

- Architected and led the development of a dynamic classifier selection (DCS-LCA) framework, implementing KNN-based overlap cleaning and expert selection across heterogeneous models (Naive Bayes, XGBoost, SVM).
- Lead the model evaluation and authored the primary manuscript, cutting log loss by 70.1% (0.898 to 0.269) versus a baseline and achieving 91.62% accuracy (macro F1: 0.8638) on the UCI Student Dropout dataset (n=4,424).

ORGANIZATIONS

Himpunan Mahasiswa Teknik Informatika (HIMTI) BINUS University

Activist in Publication and Marketing

Oct 2024 – Oct 2025

- Coordinated event planning and served as the Event Coordinator for faculty programs under the School of Computer Science (TECHNO, SESVENT, and FEAST 2025), managing end-to-end logistics and execution.

Freshmen Partner

Sept 2025 – Sept 2026

- Mentored first-year students to adapt to academic culture, guiding them through campus systems.

Freshmen Leader

Sept 2025

- Led orientation sessions and team-building activities during the Freshmen Year Program (FYP) to accelerate student campus integration.

PROJECTS

DataFlow Simulation — Simulated E-Commerce Data Platform with Event-Driven MLOps

Aug 2026

[Dashboard](#) | [GitHub](#)

- Built a monthly-batch medallion warehouse (Raw → Silver → Gold) on PostgreSQL, transforming 7 source tables into dbt-modeled marts for sales, logistics SLA, and user funnel analytics.
- Orchestrated a 6-DAG Airflow pipeline (load → dbt → Supabase sync → ML train/inference) via Docker-outside-of-Docker, gating retraining to every 6th batch instead of a fixed schedule.
- Trained XGBoost churn and conversion models with Optuna tuning, guarded by a champion/challenger registry that only promotes models beating both a baseline and the current champion.
- Shipped a Next.js + Supabase dashboard for live pipeline health, model performance, and predictions, pulling DAG status straight from the Airflow API.

NutriVision Pro — Computer Vision & MLOps Cloud Platform

Sept 2026

[Live Web App](#) | [Github](#)

- Architected and deployed a production-grade Computer Vision backend on AWS EC2 and Amazon S3, containerized via Docker & Docker Compose, serving a fine-tuned YOLOv11 instance segmentation model for real-time nutritional scanning and multi-class food item analysis.
- Automated a zero-downtime CI/CD pipeline using GitHub Actions and AWS ECR, ensuring seamless image building, registry syncing, and secure .env secret injection.
- Engineered a secure FastAPI service featuring cryptographic admin header validation (x-admin-token), dynamic ONNX model reloading, and Redis caching.
- Implemented cloud-native storage and retrieval workflows, enabling the API to fetch production-ready ONNX model weights directly from private S3 buckets on startup with built-in fallback verification and automated health-check endpoints.

TECHNICAL SKILLS

Languages: Python, SQL, C, Java, R, Dart (basic), JavaScript (basic)

Data Engineering & ETL: Apache Airflow, Medallion Architecture, dbt, PostgreSQL, Supabase, Pandas, NumPy

Model Development & Tuning: PyTorch, Scikit-Learn, XGBoost, LightGBM, Computer Vision (YOLOv11, Grounding DINO), NLP (BERT), Optuna

MLOps, Deployment & Cloud: Docker, Continuous Training Pipelines, GitHub Actions (CI/CD), AWS, Firebase, Supabase, Vercel, Git